Armed Forces University "ESPE"

Project Information

Project: SaborCode

Team: Codex

Members: Anghelo Suntaxi, Denis Paucar, Daniel Sangoquiza

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document specifies the functional requirements for SaborCode, an interactive QR-code digital menu and ordering system designed to streamline ordering, facilitate communication between diners, kitchen staff, waitstaff, and administrators, and improve overall restaurant operations.

1.2 Scope

SaborCode enables diners to scan QR codes at tables to view menus, filter dishes, and place orders directly. It includes a Kitchen Display System (KDS) for real-time order processing, stock management, digital waiter calls, and an administration module for menu and role management.

2. Overall Description

2.1 Product Perspective

The system consists of three main modules:

1. Diner Module: Mobile web interface for scanning QR codes, browsing menus, filtering dishes, placing orders, and requesting service.

2. Kitchen Module: Interface for kitchen staff to view incoming orders and update dish availability in real time.

3. Administration Module: Panel for managing categories, dishes, prices, allergen info, and user access permissions.

2.2 User Classes

• **Diner:** Customers accessing the digital menu via table QR codes.

• **Administrator:** Managers handling menu content, pricing, allergens, and system users.

• **Kitchen Staff:** Personnel viewing incoming orders and setting item stock availability.

• **Waitstaff:** Servers receiving waiter calls and bill requests on mobile devices.

3. System Functional Requirements

RF001: QR Code Scanning & Direct Redirection

User Story ID: US001 | Priority: High | Owner: Anghelo Suntaxi

The system shall allow diners to scan a table QR code to access the digital menu instantly without waiting for a printed menu.

RF002: Menu Category Management

User Story ID: US002 | Priority: High | Owner: Daniel Sangoquiza

The system shall allow administrators to manage menu categories (e.g., starters, main dishes, drinks) to keep items organized.

RF003: Dish Management & Allergen Information (CRUD)

User Story ID: US003 | Priority: High | Owner: Daniel Sangoquiza

The system shall allow administrators to add, update, and delete menu items along with pricing, descriptions, and allergen warnings.

RF004: Dish Search & Dietary Filtering

User Story ID: US004 | Priority: Medium | Owner: Daniel Sangoquiza

The system shall allow diners to filter and search dishes by category or dietary needs to easily find what they want.

RF005: Digital Ordering Workflow

User Story ID: US005 | Priority: High | Owner: Anghelo Suntaxi

The system shall allow diners to select dishes and place orders directly from their phones to the kitchen.

RF006: Kitchen Display System (KDS)

User Story ID: US006 | Priority: High | Owner: Denis Paucar

The system shall provide kitchen staff with a real-time view of incoming orders to prepare dishes in the correct sequence.

RF007: Real-Time Stock Availability Toggle

User Story ID: US007 | Priority: Medium | Owner: Anghelo Suntaxi

The system shall allow kitchen staff to mark dishes as 'sold out' in real time to prevent orders of unavailable items.

RF008: Waiter Call & Bill Request System

User Story ID: US008 | Priority: Low | Owner: Daniel Sangoquiza

The system shall allow diners to request waiter assistance or ask for the bill directly from their phone.

RF009: Role-Based Access Control (RBAC)

User Story ID: US009 | Priority: Medium | Owner: Denis Paucar

The system shall manage system users and roles (admin, kitchen, waitstaff) to control access to panel functions.

4. Requirements Traceability Matrix

User Story ID	User Story	Task Name	Task Owner	Priority
US001	As a diner, I want to scan a QR code at my table so that I can access the digital menu instantly without waiting for a printed menu.	Implement QR code scanner & redirect	Anghelo	**High**
US002	As an administrator, I want to manage menu categories (e.g., starters, main dishes, drinks) so that items remain organized for customers.	Create menu category management	Daniel	**High**

US003	As an administrator, I want to add, update, and delete menu items (with price, description, and allergen info) so that the menu stays accurate.	Implement dish CRUD & allergen info	Daniel	**High**
US004	As a diner, I want to filter and search dishes by category or dietary needs so that I can easily find what I want to eat.	Develop dish search and filtering	Daniel	**Medium**
US005	As a diner, I want to select dishes and place an order from my phone so that my order reaches the kitchen directly.	Implement digital ordering workflow	Anghelo	**High**
US006	As kitchen staff, I want to view incoming orders in real time so that I can prepare dishes in the correct order.	Create kitchen display system	Denis	**High**
US007	As kitchen staff, I want to mark dishes as "sold out" so that diners cannot order unavailable items.	Implement real-time stock availability toggle	Anghelo	**Medium**
US008	As a diner, I want to request the waiter's attention or ask for the bill from my phone so that I receive faster assistance.	Implement waiter call notification system	Daniel	**Low**
US009	As an administrator, I want to manage system users and roles (admin, kitchen, waitstaff) so that access to panel functions is controlled.	Develop role-based access control (RBAC)	Denis	**Medium**